

The Effects of Unemployment Insurance on Housing-Insecure Families*

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October 29, 2024

Abstract

The primary purpose of unemployment insurance (UI) programs is to afford beneficiaries the ability to smooth consumption over employment-related income shocks. However, little is known about whether UI prevents extreme material hardship. In this paper, we evaluate the effect of UI on one extreme outcome: homelessness. With New York State and New York City administrative data, we estimate the impact of UI eligibility on homelessness using a regression discontinuity design (RDD) that exploits a cutoff based on workers' highest quarterly earnings in the past year. We find that UI eligibility reduces homelessness within three years by 6 percentage points (a 46 percent decline). Not accounting for how UI prevents extreme distress undervalues the benefits of UI.

JEL Classification: H53, I38, J65, R28, R38

Keywords: unemployment insurance, homelessness, housing security, employment

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1 Introduction

The primary purpose of unemployment insurance (UI) programs is to afford beneficiaries the ability to smooth consumption over employment-related income shocks. The empirical work focused on documenting this component of UI has mainly considered its impact on food expenditures (Browning & Crossley, 2001; East & Kuka, 2015; Gruber, 1997). UI may play an important role in preventing many forms of extreme hardship that changes in food expenditures would not capture, such as homelessness, eviction, bankruptcy, health care use, initiation of drug or alcohol use, crime, and even death. Due to the high private and social costs of these outcomes, failing to account for how UI prevents extreme hardship may greatly underestimate its value.

In this paper, we evaluate the effect of UI in the state of New York on one extreme outcome: homelessness. Homelessness can be particularly costly to both the individual and to governments. In addition to the direct cost of providing shelter, there are increased public costs associated with homelessness stemming from higher expenditures on health care, criminal justice, and welfare programs (Flaming et al., 2015). The private cost of homelessness is very high. For example, the age-adjusted mortality rate for the homeless population in New York is four times that of the general US population (Barrow et al., 1999). A third of children in families who experienced homelessness have been separated from their families at some point in the past (Walton et al., 2018). This rate is 3-4 times higher than that of similarly poor children (Cowal et al., 2002). Beyond consumption smoothing, preventing these forms of extreme hardship may have the additional benefit of avoiding negative long-run outcomes.

Two obstacles make studying the UI/homelessness link challenging. First, UI receipt and amount are endogenous. Individuals who are eligible for or receive UI are different from others in many ways that might be correlated with homelessness. For example, some groups are over-represented among the homeless such as veterans, the mentally ill, those with a history of substance abuse, those who are HIV positive, and victims of domestic violence (Evans et al., 2021). It is highly likely these groups also are less likely to have qualifying work to become eligible for UI. Previous work demonstrates that the wage losses associated with mass layoffs start well before the layoff (Jacobson et al., 1993; Sullivan & von Wachter, 2009). These negative income shocks might affect both the generosity of UI through its effect on pre-separation earnings and the likelihood one is homeless. There may also be simultaneity bias, where being homeless impacts one's ability to access or maintain eligibility for UI. Most state UI programs require recipients to actively look for work to maintain benefits which is made more difficult by homelessness.

We overcome these endogeneity issues by using a fuzzy regression discontinuity design (RDD), exploiting an eligibility rule in New York requiring a claimant's highest-earning quarter in the last five completed

quarters to be above a certain threshold, roughly \$1900. By comparing workers who applied for UI around this cutoff, we can causally attribute any difference in homelessness across these groups to UI eligibility. A similar identification strategy was used by Leung and O’Leary (2020) to estimate the impact of UI on means-tested benefit receipt using Michigan administrative data. Differing from most papers estimating the causal effects of UI, this paper identifies a marginal effect on the extensive margin, UI eligibility and receipt, instead of the intensive margin, UI benefit amount and potential duration.

A second obstacle in identifying a causal effect of UI on homelessness is obtaining adequate data. In order to implement our identification strategy, we need claim-level UI data linked to quarterly earnings and data on homelessness. We overcome this issue by merging administrative UI data from the New York State Department of Labor (NYSDOL) to administrative data from the New York City Department of Homeless Services. An advantage of using administrative data rather than survey data is that UI is significantly and increasingly underreported in surveys (Meyer et al., 2015), and most surveys do not measure homelessness.

We find that UI eligibility reduces the probability of a homeless shelter application within a year by 3.2 percentage points (a 57 percent decline) and within three years by 6 percentage points (46 percent). These estimates are quite large. The impact of UI on homelessness appears to grow through time well beyond the expiration of UI benefits. Using rough estimates of the fiscal costs of homelessness (Flaming et al., 2015; Spellman, 2010), we find that at the minimum earnings threshold, preventing homelessness recouped roughly 20 percent of UI spending. Additionally, we find that UI eligibility increases nonemployment duration by 2.2 weeks or 6 percent. This is a small and statistically insignificant effect. However, we are unable to reject large effects such as those estimated by Leung and O’Leary (2020).

This paper contributes to four areas of research. First, it adds to research on the role of social insurance in preventing extreme material hardship. Most of the research on social insurance programs has focused primarily on labor supply effects and moral hazard. Research that estimates the impact of social insurance programs on severe material distress remains sparse. Deshpande et al. (2021) utilize age-based eligibility rules to estimate the impact of disability programs on bankruptcy, foreclosure, and home sales. They find that disability allowances reduce adverse events. Standard calculations of the optimal benefit level generally use mean consumption as a sufficient statistic for welfare gains from smoothed consumption. Incorporating their estimates, Deshpande et al. extend the standard calculation of the optimal benefit level to include tail consumption risk and negative externalities from these events. This increases the optimal benefit level.¹ In our context, we find an economically meaningful impact of UI on tail end consumption measured by homelessness. Not incorporating this tail-end outcome and the social costs associated with homelessness

¹The increase is relatively small, \$230 or roughly 2 percent. This is likely a lower bound if the authors were to consider the entire distribution of consumption because of underestimating the difference between average marginal utility and marginal utility of the average.

would underestimate the value of UI.

Our findings are also related to Hsu et al. (2018). Leveraging changes in benefit level and duration across state and time, they estimate that UI expansions in the Great Recession averted over a million foreclosures. Additionally, UI extensions insulated housing values from adverse unemployment shocks. They argue that optimal unemployment insurance frameworks (Baily, 1978; Chetty, 2006) should also consider the negative spill-overs caused by foreclosures. Similar to Hsu et al. (2018), our paper measures the impact of UI on housing stability. Some key differences are that we examine the impact of UI on a much more extreme measure of housing stability, i.e., homelessness, and the population in this analysis is primarily renters, while Hsu et al. restrict to homeowners. Additionally, we identify the effect of UI eligibility rather than benefit duration. Similar to bankruptcies and foreclosures, homelessness has large social costs.

Second, this paper contributes to the broad literature on homelessness and homelessness prevention, of which, an overview is contained in Evans et al. (2021). Much of the empirical work to date has focused on how best to intervene once people become homeless. There is very little evidence on what policies are effective in preventing homelessness. There is experimental evidence that housing vouchers prevent homelessness (Wood et al., 2008). Using seemingly random variation in the availability of funds from the Homelessness Prevention Call Center in Chicago, Evans et al. (2016) found that emergency financial assistance reduced shelter entry. As job loss is the most common reason for requesting emergency financial assistance, it seems plausible that UI and emergency financial assistance play similar roles in preventing homelessness. To our knowledge, this is the first paper that causally estimates the impact of a non-housing public social safety net program on preventing homelessness.

Although those experiencing homelessness are among the most disadvantaged in the United States, the extreme poverty literature often ignores them. This is mainly due to their absence in major household surveys such as the Current Population Survey (CPS) and the Survey of Income and Program Participation (SIPP). Recently researchers have linked administrative data to study this population. Notably, Meyer et al. (2021) has linked the population experiencing homelessness measured by the 2010 Decennial Census and the 2006-2016 American Community Survey (ACS) to administrative data on income and government program participation. They found high formal employment rates amongst both the sheltered homeless (53 percent) and the unsheltered homeless (40 percent). von Wachter et al. (2020) linked LA County Homeless Management Information System (HMIS) data with UI quarterly earnings records. They found that employment and earnings meaningfully decline before the quarter of HMIS enrollment, and then employment rebounds.

These two findings that a large fraction of those experiencing homelessness are attached to the labor market before being homeless and that earnings and employment rates drop before using homeless services presents a puzzle. Why is this population not sufficiently insured against adverse employment shocks?

It seems likely that this population is highly credit constrained, preventing self-insurance. Issues such as adverse selection and concentrated risk prevent private insurance from forming. However, why is unemployment insurance not sufficiently insuring this population? Our findings indicate that monetary eligibility requirements contribute to this lack of insurance.

Last, this paper contributes to the extensive literature on the consumption smoothing value of UI (Browning & Crossley, 2001; East & Kuka, 2015; Gruber, 1997) and the labor supply impact of UI (Johnston & Mas, 2018; Lalive, 2008; Landais, 2015; Schmieder et al., 2012, 2016). Differing from nearly all papers in this literature, our paper estimates the effect of eligibility rather than benefit amount or duration. This is relevant given that policy makers can also change parameters related to eligibility, in addition to benefit amount and duration. For example, UI eligibility was expanded to millions of displaced workers in 2020 through the Pandemic Unemployment Assistance program. Leung and O’Leary (2020) estimate the impact of UI eligibility on labor supply using a nearly identical strategy and find that UI eligibility increases nonemployment duration by 7.4 weeks (a 23 percent increase). Our estimates are considerably smaller. We find that UI eligibility increases non employment duration by roughly 2.2 weeks (a 6 percent increase). However, our estimates are imprecise, and we can only reject effects larger than 9.4 weeks.

Previous work has shown that the impact of UI on consumption smoothing varies significantly across individuals. In particular, Browning and Crossley (2001) show that this benefit is larger for those without liquid assets. The estimated impact of UI on consumption smoothing is driven by ten percent of their sample, where there is a one-to-one relationship between UI benefit level and total expenditures. Since it seems likely that the population in our paper does not have significant assets and is credit constrained, our results are consistent with UI being particularly important for credit-constrained individuals. If we take the change in mean income as an upper bound for change in mean consumption, we show that small changes in mean consumption can mask large changes in consumption in the tails of the distribution as measured by homelessness. This finding cautions against using just mean consumption change to calibrate optimal benefit levels (Baily, 1978; Chetty, 2006).

2 Institutional Background

The United States unemployment insurance (UI) system provides temporary and partial wage replacements to eligible workers that lose their job. Although federally mandated, the UI system is state-run. States have the power to set program parameters such as benefit amount and eligibility. Although the exact structure of UI varies from state to state, the broad structure remains relatively similar. This section focuses on the UI institutional details of New York state over the relevant period for this study. New York’s UI system is

largely representative of the US as a whole.

A claimant must satisfy both the “monetary” and “non-monetary” eligibility criteria to qualify for UI benefits. The monetary eligibility criteria require that a worker have sufficient labor force attachment prior to the job separation. Specifically, wages in the “base period,” either the first four of the last five completed quarters (standard base period) or the four most recently completed quarters (alternative base period), must satisfy two criteria. First, the highest-earning quarter in the base period must be about \$1,900, depending on the year.² Second, total earnings in the base period must be at least 1.5 times higher than the highest quarter earnings. Compared to other states, New York has relatively strict monetary eligibility requirements (Leung & O’Leary, 2020). In addition to monetary eligibility, a worker must be non-monetarily eligible, meaning the worker must not have quit or been fired for cause.

The weekly benefit amount (WBA) is calculated from a claimant’s highest-earning quarter. Figure 1 displays the weekly benefit schedule from 2004 to 2013 that maps the highest-earning quarter to WBA. For most of the sample period and those barely above the minimum earnings threshold, WBA is 1/25 of the highest-earning quarter. This amount is roughly a 50 percent replacement rate. If there is variability in a claimant’s quarterly earnings, the replacement would be much higher because it is calculated from the highest-earning quarter, not average quarterly earnings. For 2015 and onwards, there is a \$100 WBA minimum. For a claimant just above the highest-earning quarter cutoff, this minimum causes the replacement rate to be around 68 percent. At a higher threshold, \$3,575, WBA is 1/26 of the highest earnings quarter. This change results in roughly \$6 discontinuity. WBA is capped at roughly \$400, so a highest-earning quarter would have to be around \$11,000 to be above this cutoff.

New York, along with eight other states, is a uniform duration state. This means that workers eligible for UI are entitled to 26 weeks as long as they remain unemployed. Our sample period encompasses the Great Recession, where the Extended Benefits (EB) and Emergency Unemployment Compensation (EUC08) programs dramatically expanded UI duration. Some claimants were eligible for 73 more weeks, 99 weeks in total. Additionally, the American Recovery and Reinvestment Act of 2009 increased weekly benefit levels by a flat \$25, a 40 percent increase for claimants at the highest-earning quarter threshold, for all recipients from March 2009 through May 2010.

²The cutoff is indexed to the state’s minimum wage. Specifically, it is 221 times the states minimum wage rounded to the lowest \$100. The cutoff was nominally \$1,600 from 2004 to 2013. Then it gradually increases to \$1,900 by 2016.

3 Data and Empirical Strategy

3.1 Data

Our data are from administrative records from New York City and the New York State Department of Labor. The composition of our sample is dictated, in part, by the initial use of this data, examining the effects of evictions (Collinson et al., 2024). We begin with administrative data on historical benefits receipt from New York City’s Human Resources Administration (HRA). These records include individuals who received Medicaid, Temporary Assistance for Needy Families (TANF), Supplemental Nutrition Assistance Program (SNAP; food stamps), or other city-specific cash subsidies between 2004 to 2016.^{3,4} From this point, there are two samples constructed. The first sample we call the *courts* sample. This sample includes individuals in the HRA records that can be matched to the universe of housing court records in New York City from 2007-2016. These records are mostly eviction filings. Roughly 40 percent of these court records can be matched to HRA benefit receipt data. The matching procedure is described in Collinson et al. (2024). The second sample is a random subsample of individuals who appear in the HRA benefits records but are not linked to housing court records. We call it the *non-courts* sample.

Using social security numbers, we link our sample to quarterly earnings data from the New York State Department of Labor (NYSDOL) from 2004-2016. These data include detailed industry codes (six-digit NAICS) and cover approximately 97 percent of New York State’s non-farm employment. However, these data do not capture private household workers, student workers, the self-employed, or unpaid family workers. In addition to quarterly earnings data, we link our samples to claim-level unemployment insurance data from 2004-2016. This data includes the date when a claim was made, whether the claim was deemed eligible, the total benefit amount received, the number of weeks receiving UI, and basic demographic information. This data includes both regular state UI and UI paid through EB and EUC08 programs.

To estimate the effect of UI on homelessness, we link our samples to administrative data covering applications to homeless shelters from the New York City Department of Homeless Services from 2003 to 2017. These data contain information on applications to and stays in New York City homeless shelters. To measure homelessness, we focus on applications to shelters rather than stays. Individuals who apply to stay in a homeless shelter but are deemed ineligible are almost certainly experiencing homelessness.

We make a couple of other restrictions on our samples. First, we exclude claims made within two years of prior claims. This restriction ensures that each UI claim is a new unemployment spell. We observe a mass of claims a year after the initial claim. Since a claim is only valid for one year, a new claim must be filed

³This data does not capture Medicaid clients receiving Medicaid from the state Department of Health.

⁴Benefit receipt must occur prior to an eviction filing.

to continue receiving benefits from the EB or EUC08 programs. Second, we restrict to claims satisfying the other monetary requirement, total earnings in the base period must be 1.5 times greater than earnings in the highest-earning quarter. Third, we restrict to individuals aged 18 to 65 and exclude individuals living in public housing (New York City Housing Authority). Because the New York City Housing Authority is very lenient regarding rent repayment, employment shocks are unlikely to cause homelessness for these individuals. Finally, we restrict to claims where the highest-earning quarter of the last five completed quarters is within \$1,500 of the minimum earnings threshold. After these restrictions, we are left with 41,320 claims in the *courts* sample and 12,229 claims in the *non-courts* sample. In our main analysis, we combine both of these samples.

Table 1 presents descriptive statistics for new claims without restricting to claims around the cutoff (columns 1 and 2) and our main sample (column 3). Columns 2 and 4 show descriptive statistics from the *non-courts* sample, and column 5 shows descriptive statistics for the re-weighted sample. Since our sample is comprised of UI claims from means-test benefit recipients, the individuals in our sample are relatively disadvantaged, even before restricting to claims around the cutoff. For example, the mean yearly earnings for this population is \$26,523, and nearly 13 percent have applied to stay in a homeless shelter prior to filing a UI claim. As a whole, this population is disproportionately female, Black, Hispanic, and less educated.

After restricting to claims around the highest-earning cutoff, the sample is even more disadvantaged. The previous year’s annual earnings are roughly \$7,000. This is well below the federal poverty line. One-fifth of this sample has previously applied to stay in a homeless shelter. A third of the sample received TANF within two quarters before the UI claim. For reference, less than one percent of US families receive TANF. Around 70 percent of this sample is female, and 44 percent of this sample is Black. Less than 5 percent of this sample has a Bachelor’s degree.

By combining the *courts* and *non-courts* sample, we are overweighting individuals who have an eviction filing linked to them. The individuals in the *courts* sample are relatively disadvantaged compared to those in the *non-courts* sample. For example, only 10 percent of individuals in the *non-courts* sample have applied to stay in a homeless shelter relative to 20 percent in the combined sample.

In summary, individuals in our sample are extremely disadvantaged due to four reasons: all individuals in our sample have historically received a means-tested benefit between 2004-2016, all individuals have recently lost employment, all individuals are within \$1,500 of an approximately \$1,900 highest-earning quarter threshold, and our sample overweights individuals with previous or future eviction filing linked to them. Since having an eviction filing is likely endogenously related to receiving UI, one concern is that we have endogenous selection into our sample. In section 5.1, we mitigate this concern by showing that our results are robust to reweighting the sample. The restriction that all individuals in our sample have historically received means-

tested benefits from New York City’s Human Resources Administration may limit generalizability. However, around the minimum earnings threshold, most individuals should qualify for a means-tested program such as SNAP. We hope to obtain a more representative sample, such as the universe of UI claims from individuals in New York City in future work.

Throughout most of the paper, we use entitled for UI or eligible for UI as the endogenous variable. As shown in Table 1, there is a small difference between receipt and eligibility. To appear in our sample, an individual must submit a UI claim. Conditional on applying for UI, there is a 96 percent take-up rate in our main sample (column 3), calculated by dividing UI receipt by entitled to UI. In our main sample, around a fifth of claimants have previously applied for UI benefits.⁵ In the main sample, around two-thirds of claims are deemed eligible. 87 percent of claims are monetarily eligible, meaning the highest-earning quarter is above the cutoff. Conditional on eligibility, the average benefit amount received is \$3,277. Because of UI extensions through the EB and EUC08 programs, conditional on eligibility, the average number of weeks receiving UI, 30 weeks, is well above the maximum duration in regular times, 26 weeks.

3.2 Empirical Strategy

To estimate the causal impact of UI, we implement a regression discontinuity (RD) design, exploiting the criterion where a claimant’s highest quarter must be greater than a certain threshold.⁶ Leung and O’Leary (2020) use a nearly identical empirical strategy to identify the effect of UI eligibility on labor earnings and means-tested program receipt.⁷ Intuitively, we compare outcomes of individuals with similar high quarter earnings, but one group is barely eligible for UI, and the other group is barely ineligible. Any difference between the two groups in subsequent outcomes can be attributable to UI eligibility. Since there is not perfect compliance, we have to scale the difference by the increase in UI eligibility due to crossing the threshold.

We estimate the following equation,

$$Y_i = \alpha_0 + \beta_y T_i + \alpha_1 (R_i - c_t) + \alpha_2 T_i \times (R_i - c_t) + \Phi X_i + \mu_t + \epsilon_i, \quad (1)$$

using observations within h_- below the threshold and h_+ above the threshold. Y_i is an outcome variable of interest such as whether a claimant applied to stay in a homeless shelter; R_i is the running variable, the claimant’s high quarter earnings;⁸ c_t is the minimum high quarter earnings threshold needed to qualify for

⁵As described above, this excludes claims filled within two years of each other.

⁶Technically, the cutoff used is whether the maximum quarter earnings of the the last five completed quarters is above the highest quarter earnings threshold. Whether the claimant qualifies using the standard base period or the alternative base period this condition must be satisfied to qualify for UI

⁷They find that UI eligibility increases the duration of nonemployment. Additionally, UI eligibility decreases TANF participation in half; SNAP and Medicaid receipt is not affected. Overall eligible workers have 55 percent more income than ineligible workers.

⁸ R_i equals the maximum quarter earnings of the last five completed quarters.

UI; X_i is a vector of covariates; μ_t is a quarter-year fixed effect; T_i indicates whether the running variable is above the cutoff. To estimate the impact of eligibility instead of the reduced-form coefficient, we estimate

$$Y_i = \gamma_0 + \beta_{UI} UI_i + \gamma_1(R_i - c_t) + \gamma_2 T_i \times (R_i - c_t) + \Omega X_i + \theta_t + u_i, \quad (2)$$

instrument for UI eligibility, UI_i , with T_i .

For the main estimates, we use the method proposed by Calonico et al. (2014) to select the bandwidths h_- and h_+ .⁹ Intuitively, there is a bias-variance trade-off. Selecting a bandwidth too large may yield biased estimates; on the other hand, a bandwidth too small excludes too many observations creating imprecise estimates. We allow the bandwidths to vary on different sides of the threshold. We use a triangular kernel. In section 5.1, we show that our main results are consistent across different specifications.

This approach allows us to estimate a “local average treatment effect (LATE)” that is relevant for the compliers around the threshold. Even with perfect compliance, e.i., being above threshold perfectly determines UI eligibility, the average treatment effect estimated is specific to the population around the threshold. However, the reduced-form estimate is relevant for policymakers because one of the main UI policy parameters a state can set is the minimum highest quarter earnings needed to qualify for UI.

3.3 Validity of Identifying Assumptions

The key identifying assumption is that the conditional expectation function of the outcome variables is continuous around the eligibility threshold for both groups above and below the cutoff. Practically, it means that individuals barely below and above the threshold are not systematically different from another. This is often violated when agents can sort around the threshold. One may be concerned that workers or employers time the jobs separation in order to be barely eligible or ineligible for UI. Alternatively, employers may strategically pay employees just below the high quarter earnings threshold to prevent workers from claiming UI, leading to paying increased taxes.¹⁰

Figure 2 shows the density of UI claims around the threshold. We find a small jump in the density of UI claims. Formally, we implement a test of smoothness proposed by McCrary (2008) and reject the null hypothesis of no discontinuity in the density. Visually, there does not appear to be a decrease in density below the threshold. This is more consistent with individuals above the threshold being more likely to apply for UI rather than manipulating the running variable. In section 5.1, we show that our main results are robust to removing observations near the threshold, e.i., “donuts.”

⁹We use the “rdrobust” package developed by Calonico et al. (2014)

¹⁰Unemployment insurance is imperfectly experience rated.

To assess the validity of this RDD, we test whether predetermined covariates are smooth around the threshold. Table 2 presents estimates of discontinuities in predetermined covariates around the threshold, such as demographic information, prior earning, and prior homelessness. Across each covariate, no point estimate is statistically significant at the ten percent confidence level. Prior homelessness is strongly predictive of future homeless. We estimate as statistically insignificant and a less than one percentage point discontinuity in prior homelessness. In order to summarize these estimates, we linearly predict our main outcome variables, whether an individual applied to stay in a homeless shelter within three years and employment. These measures should be smooth around the threshold if individuals are not systematically different around the threshold. Figure 3 plots binned averages of these predicted outcomes, prior homelessness, and quarterly earnings two years prior to the UI claim. From visual inspection, they appear to be smooth around the threshold, and the estimated discontinuities are small and insignificant.

Leung and O’Leary (2020) estimate an RDD using a similar threshold with the universe of Michigan UI claims from 2005-2010. They do not find a jump in density around the threshold. Perhaps, in New York, it is easier to determine whether one is eligible for UI before applying, so fewer individuals apply for UI below the cutoff. Potentially, means-tested benefit recipients are more knowledgeable and more likely to apply for UI when they are just above the threshold. One concern is that we have sample-selection bias, where “knowledgeable” individuals know their eligibility and only claim UI above the threshold. Hence, they only appear in our sample above the threshold. This is a threat to identification if being knowledgeable is correlated with applying to stay at a shelter. The covariate balance provides suggestive evidence that being knowledgeable is not correlated with applying to stay at a shelter.

4 Results

4.1 UI Eligibility and Benefit Amount (First Stage)

Figure 4a plots the probability of whether a claim is deemed eligible around the highest-earning quarter threshold. There is a slightly less than 60 percentage point increase in eligibility at the threshold. Above the threshold, a claim may still be ineligible for nonmonetary reasons. For example, the claimant may have been fired for cause or quit. Below the cutoff, roughly ten percent of claims are deemed eligible. Based on New York State’s monetary eligibility rules, no claimant below the threshold should qualify for UI. One explanation for this difference is that the claimant’s employer inaccurately reported earnings. If a claimant’s earnings were underreported, the claimant may request that her earnings be reverified. Additional earnings would be added to the highest-earning quarter, but the quarterly earnings file may be left uncorrected.

Throughout this paper, the main endogenous variable is whether a claim is deemed eligible for UI. There is a small difference between eligibility and receipt. Table 3 shows estimates of equation (1) with UI eligibility, receipt, and the benefit amount received as the outcome variable. Conditional on applying for UI, take-up is 95 percent. A small number of individuals who apply for UI are deemed eligible but never received any benefit. Presumably, these individuals find employment before receiving UI payments. Because the offer of UI benefits regardless of take-up may affect key outcomes of interest such as homelessness or labor supply, using UI eligibility as the main endogenous variable captures the full effect of UI.¹¹

Figure 4b plots the benefit amount received by each claim around the cutoff. There is around a \$1300 increase in the total benefit amount received. Above the threshold, there is a positive linear relationship between high quarter wages and the benefit amount received. In New York, the weekly benefit amount is calculated by multiplying high quarter wages by 1/25. Conditional on eligibility, those just above the threshold receive \$2,219 in unemployment benefits. Although we do not have the total benefit amount a claimant is eligible for, this amount must be greater than \$2,219 because some claimants find employment before they exhaust their UI benefits.

4.2 Homelessness

In this section, we present visual evidence and formal estimates of the effect of UI eligibility on our main outcome of interest, homelessness. We measure homelessness by application to a homeless shelter. Figure 5 plots the fraction of UI claimants who apply to a shelter within 36 months after their UI claim. There is a clear drop of around 3 percentage points at the cutoff. Lower earnings are associated with a higher probability of being homeless within 36 months. Below the threshold, the curved relationship between earnings and homelessness may be explained by individuals far away from the threshold actually receiving more than \$1000 in UI benefits, as shown in Figure 4b.

Table 4, columns 5 and 6 show the drop in fraction homeless scaled by the first stage to estimate the impact of UI eligibility on homelessness. We find that UI eligibility decreases the probability of a homeless shelter application within 36 months by 6 percentage points off a control mean of 13 percentage points. This impact is statistically significant at the five percent confidence level.

Table 4 presents estimates for the effect of UI eligibility on homelessness within one, two, and three years. The even columns include controls, and odd columns exclude controls.¹² UI eligibility reduces shelter applications by 3 percentage points within one year and 4 percentage points within two years. In terms of percent decrease, UI eligibility decreases shelter applications by 57 percent within one year, 41 percent

¹¹Table 7 presents our main estimates using UI receipt and benefit amount received in thousands as the endogenous variable.

¹²All regressions include quarter-year fixed effects.

within two years, and 46 percent within three years. Validating the empirical design, the inclusion of controls leaves estimates largely unchanged. Figure 6 plots the impact of UI eligibility on the probability of shelter application cumulatively through time. The estimated effect grows through time, well beyond the duration of UI benefits. In normal times the maximum UI duration is 26 weeks or two quarters. As previously noted, our sample encompasses the 2008 financial crisis where UI was expanded up to 99 weeks, roughly two years. Conditional on eligibility, the average duration of benefits in our sample is 31 weeks. This suggests a large lag between when UI benefits are received and when homelessness is averted.

4.3 Income and Labor Earnings

It is well documented that increases in UI duration and generosity increase unemployment duration (Krueger & Meyer, 2002; Schmieder & Von Wachter, 2016). In this section, we estimate the impact of UI eligibility on measures of unemployment duration. Our two measures of unemployment duration are whether an individual is employed in the quarter after the layoff and the number of consecutive quarters with zero earnings. Figure 7 plots binned averages of these measures around the cutoff. There appears to be a decrease in the probability of employment in the quarter after the UI claim and an increase in nonemployment duration at the threshold. However, these discontinuities are statistically insignificant. Table 5 presents formal estimates of the impact of UI eligibility on these measures of unemployment duration. UI eligibility increases nonemployment duration by 0.171 quarters or 2.2 weeks. This is less than a 6 percent increase. However, this effect is imprecise. We fail to rule out large decreases, such as a 5.5 week reduction, or large increases, such as a 9.4 week increase in nonemployment duration. Using a similar threshold in Michigan and a similar time period, Leung and O’Leary (2020) find that UI eligibility increase nonemployment duration by 7.4 weeks. Section 5.1 shows that reweighting our sample to account for individuals with an eviction filing being overrepresented makes our estimate approximate Leung and O’Leary’s estimate.

Turning from employment to earnings, we plot the estimated impact of UI eligibility on earnings for each quarter within three years of the claim date in Figure 8a. In the quarter of the claim, there is no difference in earnings. For the next three quarters, earnings are lower for those eligible for UI, although this effect is only statistically significant for the quarter after the UI claim. From four to nine quarters after the claim, earnings are higher for those eligible for UI, but this difference is statistically insignificant.

Although in some quarters UI eligibility decreases earnings, this decrease may be offset by UI benefits. We assume that income is only composed of earnings and UI benefits to measure how income changes through time. Previous work has shown that UI eligibility modestly decreases means-tested benefit receipt for those around the high quarter threshold (Leung & O’Leary, 2020), so this assumption modestly overestimates

the difference in earnings between those eligible for UI and those ineligible. We distribute the total benefit amount received across each quarter using the number of weeks each claim receives benefits. Figure 8b plots the estimated impact of UI eligibility on income each quarter within three years of the claim date. In the quarter of the UI claim, being eligible for UI increases income by \$320. In the quarter after the claim, the UI benefit amount offsets slightly more than the earnings lost. The difference between the estimates in Figures 8b and 8a is the UI benefit amount in that quarter. Within two years of the claim, income is higher in every quarter for those eligible for UI. Cumulatively over the three years, income is \$1,566 greater for those eligible for UI. Around \$100 of this difference is attributable to higher earnings over three years of UI-eligible individuals.

Assuming that change in income is an upper bound for change in consumption, Figure 8b shows relatively modest changes in mean consumption. Section 4.2 shows that UI eligibility dramatically reduces homelessness, which is likely a tail-end outcome. Together, these two results suggest that mean consumption changes can often hide consumption changes in the distribution’s tails. Because most utility functions are assumed to be concave, reducing the variability of consumption while holding the mean constant will increase mean utility. Because UI likely reduces the variability in consumption, using increases in mean consumption will underestimate the consumption smoothing value of UI.

5 Robustness and Alternative Specifications

5.1 Robustness

In this section, we show how our estimates change to a variety of robustness checks. Table 6 presents estimates using donuts, different bandwidths, a different kernel, higher degree polynomials, and reweighting the sample. Column 1 displays our preferred specification. Our results are largely robust to omitting observations near the threshold, e.i., donuts. The estimated impact of UI eligibility on homelessness within 36 months remains essentially unchanged using \$10 and \$25 donuts. As we increase the donut size, the standard errors increase. Additionally, more extrapolation is needed, which increases the potential bias. The main estimates on homelessness are no longer statistically significant using \$50, \$100, and \$200. As noted in section 3.3, there is a small increase in density at the threshold. The robustness of our main estimates to donuts suggests that manipulating the running variable is not a threat to identification.

Columns 7 and 8 contain our main estimates using a mean-squared optimal bandwidth requiring the bandwidth to be the same on both sides of the threshold. This restriction produces slightly smaller estimated reductions of homelessness. These estimates are no longer statically significant; however, our preferred

estimates are still within a standard error of the estimates in columns 7 and 8. Given that the density of observations increases as high quarter wages increase, and generally, there is more curvature to the left of the threshold, allowing bandwidths to vary on either side of the threshold seems reasonable. No estimates appear to meaningfully change by using a uniform kernel instead of a triangular kernel. Additionally, the degree of the polynomial used to estimate the discontinuities appears not to matter.

As noted in section 3.1, our main sample combines the universe of claims made by means-tested benefit recipients in New York with eviction filing linked to them and claims from a random subsample of means-tested benefit recipients without an eviction filing linked to them. This means we are overweighting claimants an eviction filing linked to them. This population is more housing insecure. Since having an eviction is likely endogenous to receiving UI, we may have sample-selection bias. We can correct this by up-weighting claimants from the *non-courts* sample. Column 13 shows our main estimates with a reweighted sample. Surprisingly, this increases our estimates of UI eligibility on homelessness even as the population mean decreases. Additionally, the estimated effect of UI eligibility on nonemployment duration increases to 0.508 and is statistically significant. This estimate is much closer to Leung and O’Leary’s (2020) of 0.57, who use a nearly identical identification strategy.

5.2 Alternative Thresholds

As noted in section 2, there are two requirements needed to be monetarily eligible for UI. First, a claimant’s highest-earning quarter in the base period must be greater than a certain threshold (roughly \$1,900). Second, total wages in the base period must be 1.5 times greater than the highest-earning quarter. Most of this paper uses the first requirement. In this section, we use the other requirement to calculate similar RDD estimates. These estimates are shown in column 1 of Table 8. The first stage in terms of eligibility is roughly 12 percentage points smaller. Covariates appear to be balanced across this threshold. Using this threshold, the estimated effect of UI eligibility on homeless shelter application is positive, small, and insignificant. The estimated effect on nonemployment duration is slightly larger but still insignificant.

By taking the minimum of both running variables, we can use both cutoffs to estimate an RDD. These estimates are shown in column 2. The estimates of the effect of UI on homelessness are negative, small, and insignificant. One explanation for the differences between using the high quarter cutoff and the base period cutoff RDDs is that claimants around the base period threshold are slightly better off.

6 Conclusion

This paper explores whether UI prevents extreme material hardship. In particular, we estimate the impact of UI eligibility on homelessness measured by applications to homeless shelters. Using an RDD, we find that UI eligibility reduces the probability of a homeless shelter application by 3.2 percentage points within one year and 6 percentage points within three years. These effects are quite large, roughly reducing homelessness by half. UI eligibility continues to prevent homelessness well beyond the expiration of benefits. These estimates are robust to a variety of specifications.

Spellman (2010) estimates \$2,400 as the average per-person cost of housing individuals who become homeless for the first time. Including other costs such as health care, criminal justice costs, and the use of welfare programs, one study estimates the per-person public cost of the homeless to be \$5,148 (Flaming et al., 2015). These are rough estimates and likely affected by omitted-variable bias. However, they may serve as a reasonable upper bound. Using these estimates, UI recoups roughly 20 percent of UI spending through homelessness prevention at the minimum earnings threshold. In addition to fiscal savings, most of the benefits of preventing homelessness accrue to the individual through reduced hardship, such as decreased mortality risk (Barrow et al., 1999). Given that UI eligibility significantly reduces homelessness, using changes in consumption measured by food expenditures to estimate the benefit of UI will undervalue UI. Optimal UI design (Baily, 1978; Chetty, 2006) should consider how UI prevents extreme hardship, in addition to consumption smoothing and the distortionary impact on labor supply.

Our results reveal a significant gap in the social safety net. In 2019, over 760 thousand UI claims (10 percent of all claims) were denied due to insufficient prior earnings. Our results suggest that this population is at risk for extreme material hardship. Previous work has shown that only a small portion of the UI benefit denied is made up through other social safety net programs (Leung & O’Leary, 2020). Insuring this population from extreme hardship would be relatively cheap, roughly \$2,217 per person or \$1,774, including the fiscal savings from preventing homelessness. Holding fixed the number of claims and extrapolating with our estimates, removing monetary eligibility requirements would cost \$775.2 million or a 3 percent increase in total UI outlays. Relative to increasing duration or benefit amount, our estimates are suggestive that expanding eligibility better targets relief to those likely to experience extreme hardship. Prior proposed legislation planned to lower the highest-earning quarter to \$1,000 nationally.¹³ We estimate that this may dramatically reduce the risk of homeless that low-income displaced workers face. This benefit should be weighed against other well-documented costs of expanding UI, such as the distortionary effect on labor supply. This paper focuses exclusively on monetary eligibility. Less is known about the impact of expanding

¹³U.S. Senate Committee on Finance, “Wyden, Bennet, Brown introduce bill to provide down payment on unemployment insurance reform,” Sept. 27, 2021, <https://bit.ly/30iUVvu>.

non-monetary eligibility and increasing take-up.

In this paper, we examine the impact of UI eligibility on homelessness. Homelessness is an extreme form of housing instability. In future work, we hope to measure the impact of UI on other indicators of housing instability, such as moves measured by commercial reference data (Phillips, 2020) and evictions. Additionally, we plan to estimate the impact of UI on financial health measured with credit data. As noted above, the estimates used to construct the fiscal cost of homelessness are rough and may not be causal. In future work, we hope to directly measure these costs, such as increased emergency room use. Given the negative impact of job loss on mortality (Sullivan & von Wachter, 2009) and higher mortality rates among the homeless (Barrow et al., 1999), it seems plausible that UI eligibility may reduce mortality. UI and emergency financial assistance likely play similar roles in preventing extreme hardship. Palmer et al. (2019) find that emergency financial assistance reduces violent crime by 51 percent. Jenkins (2024) provides evidence suggesting that UI eligibility leads to a short-term decrease in arrests. Lastly, instability caused by adverse employment shocks may have intergenerational impacts. We hope to estimate how UI mitigates these impacts by linking similar UI claim data to schooling outcomes.

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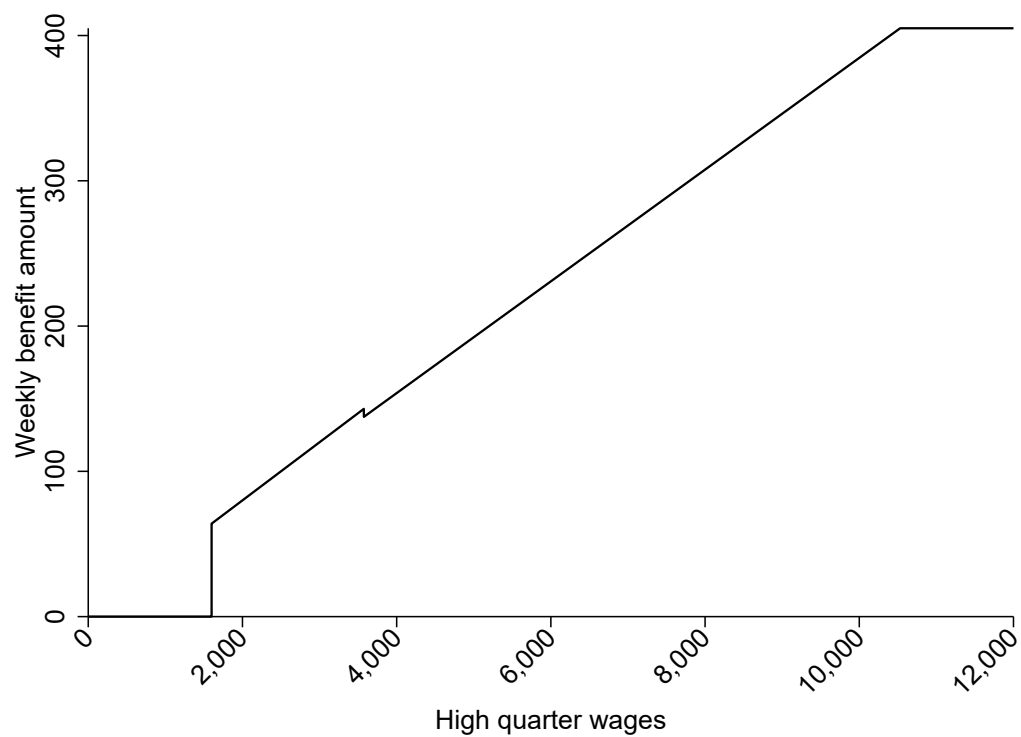
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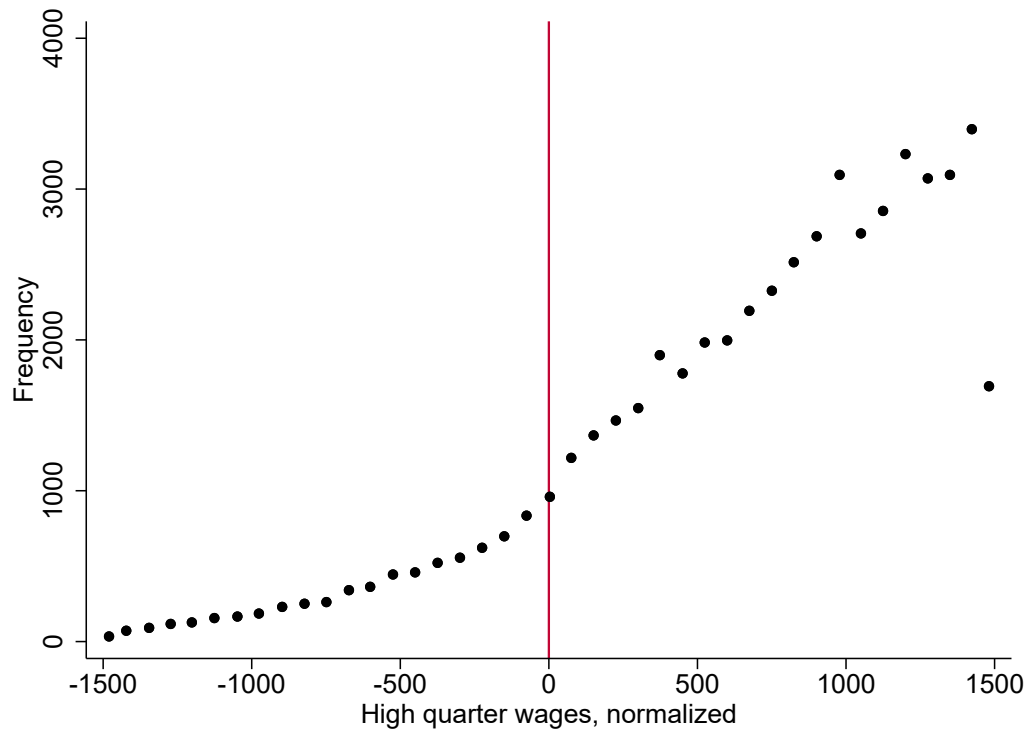
7 Figures

Figure 1: 2004-2013 Weekly Benefit Amount Schedule



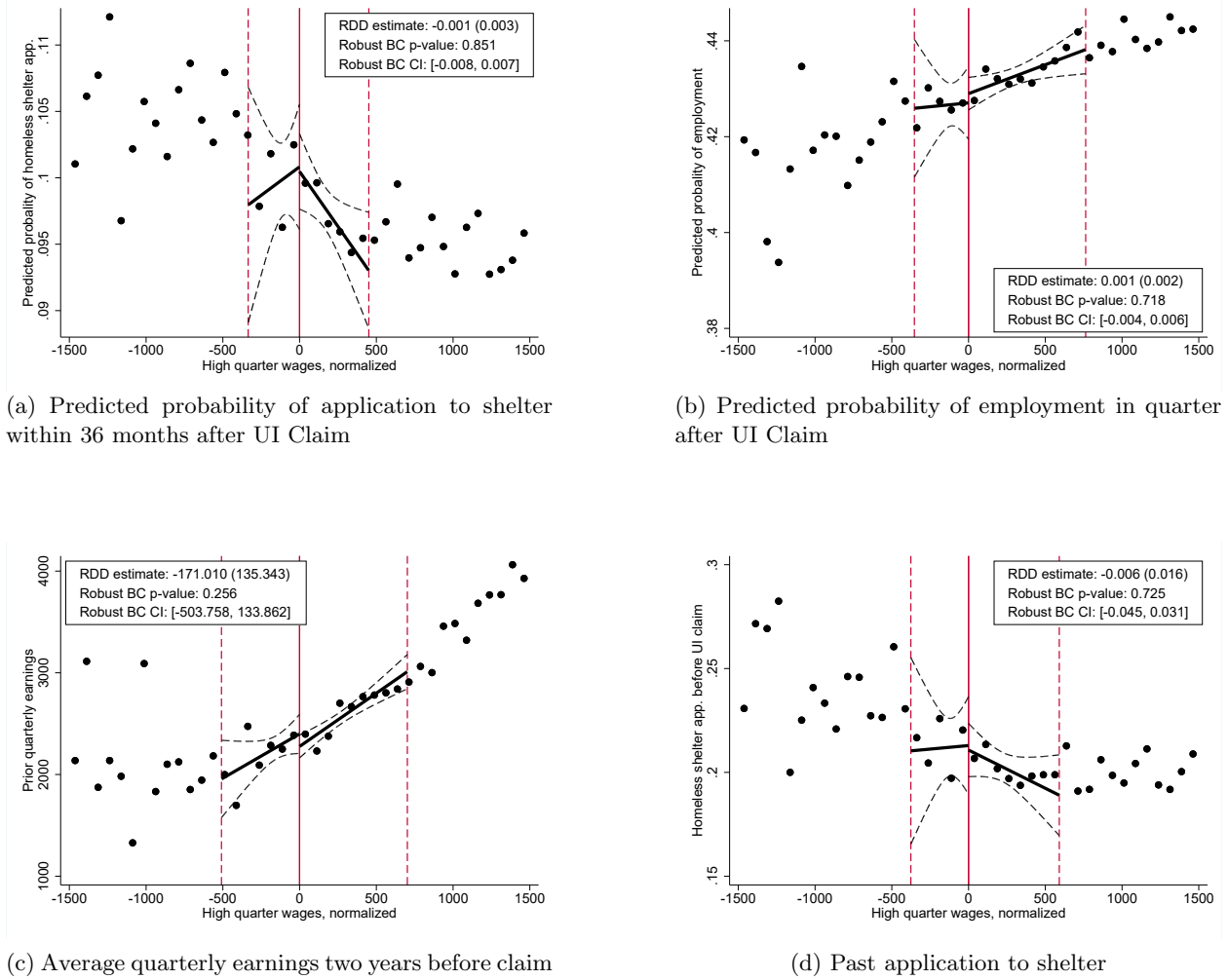
Notes: This figure plots the weekly benefit amount schedule for New York State from 2004-2013. Source: United States Department of Labor <https://oui.doleta.gov/unemploy/statelaws.asp>

Figure 2: Density of Claims around Eligibility Threshold



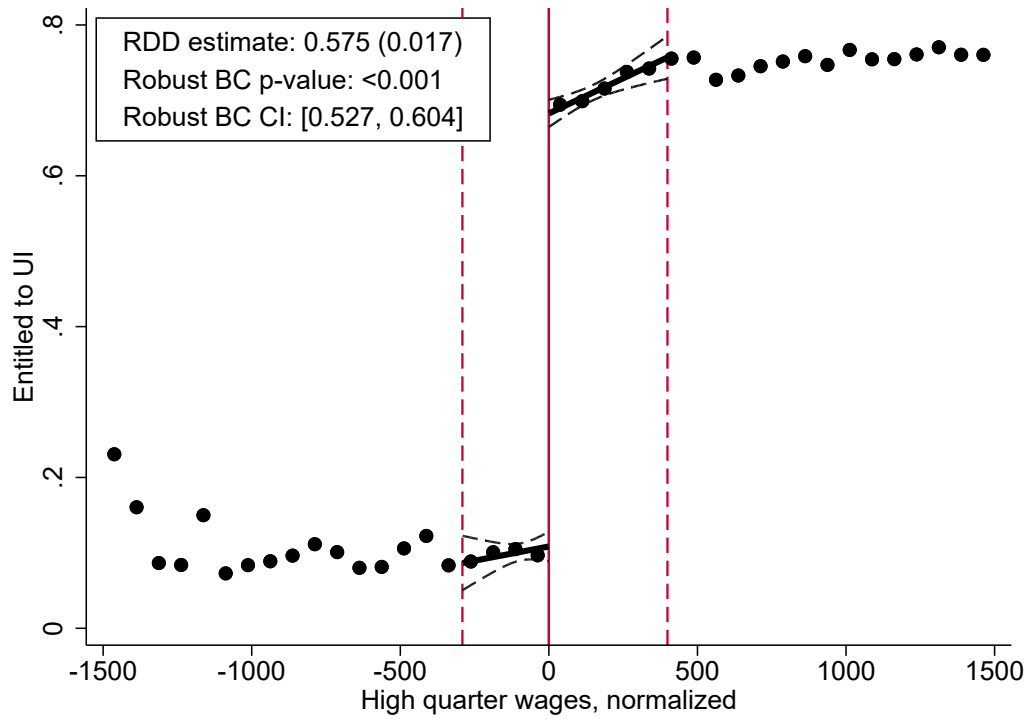
Notes: This figure plots the number of UI claimants in each nonoverlapping \$75 interval of (normalized) high quarter earnings. The vertical line denotes the minimum earnings threshold.

Figure 3: Covariate Balance Around Eligibility Threshold

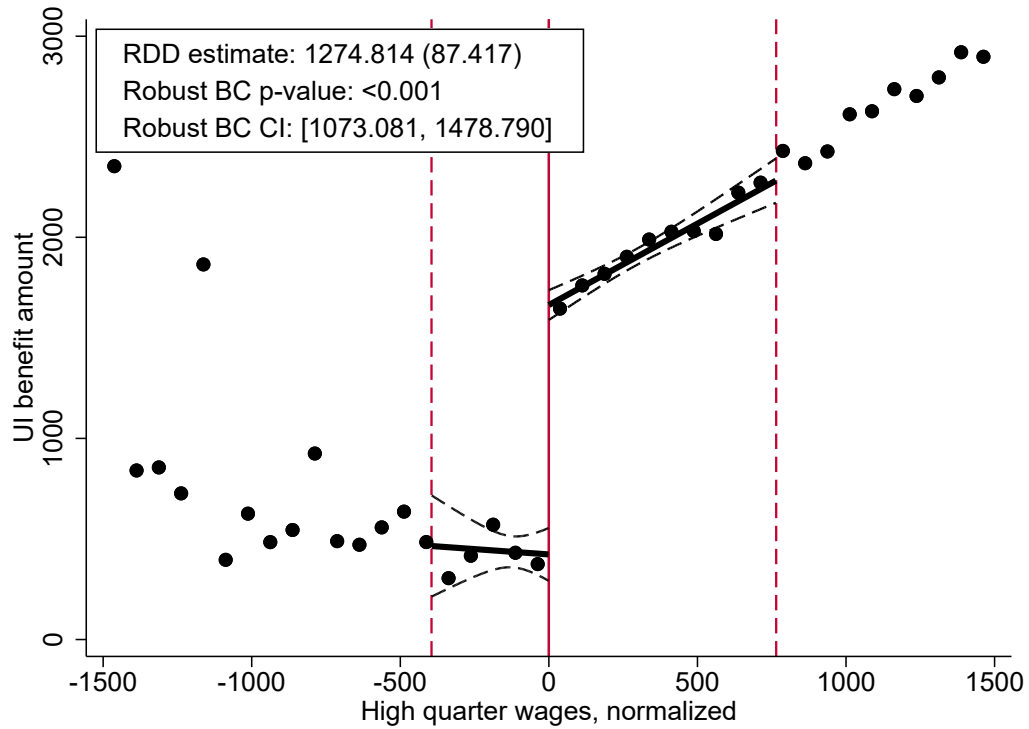


Notes: These figures summarize the relationship between observable predetermined characteristics and the running variable of UI claimants. The summary indexes in panel (a) and (b) are the predicted probability of DOC booking within 36 months and nonemployment duration after layoff is calculated using a probit model for probability of DOC booking and a linear regression for nonemployment duration with demographic information, prior earnings and employment, prior homelessness, zip code, date, and past DOC interaction. For each summary index, the mean value is plotted. Panel (c) plots the average quarterly earnings from 6-9 quarters before the UI claim. Panel (d) plots the proportion of claimants who appear in the DOC data prior to the UI claim. The vertical line denotes the minimum earnings threshold. Each dot is a nonoverlapping \$75 interval of (normalized) high quarter earnings. The dashed vertical lines denote the bandwidths. The thick solid lines within the bandwidths are the local linear fits, and the gap between them at the cutoff date equals the RDD point estimate. The dashed lines around the local linear fits are 95 percent confidence intervals. In the figure, I report the RDD point estimate, the conventional standard in parentheses, and the robust bias-corrected p-value. All other RDD figures in this paper use the same methodology.

Figure 4: UI Eligibility and Benefit Amount (First Stage)



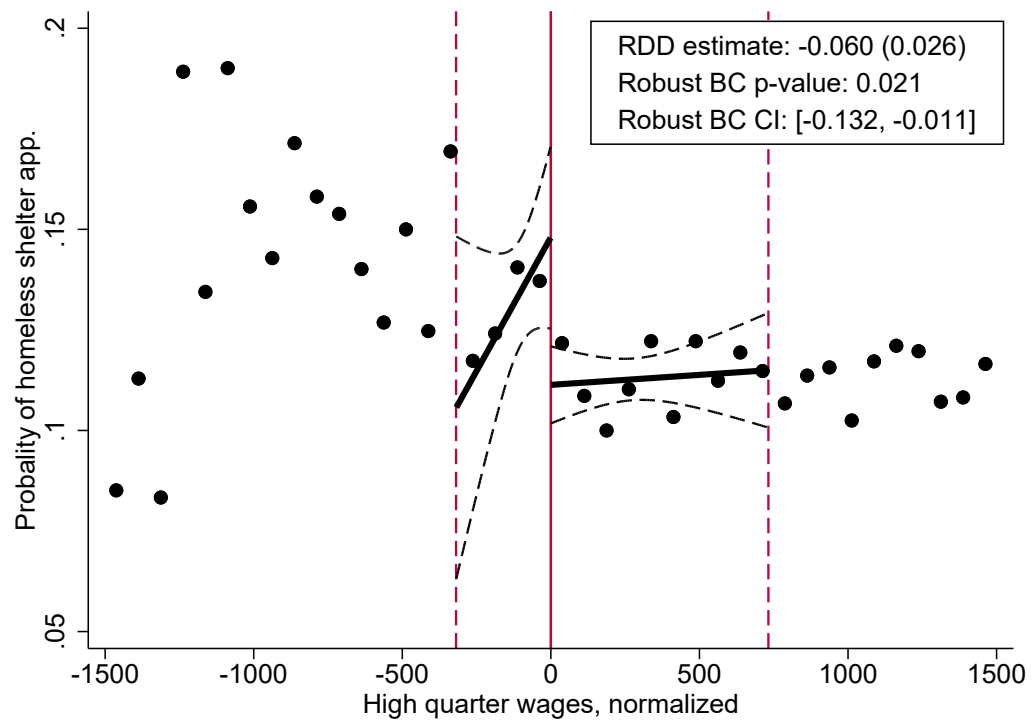
(a) Probability of UI eligibility



(b) UI benefit amount

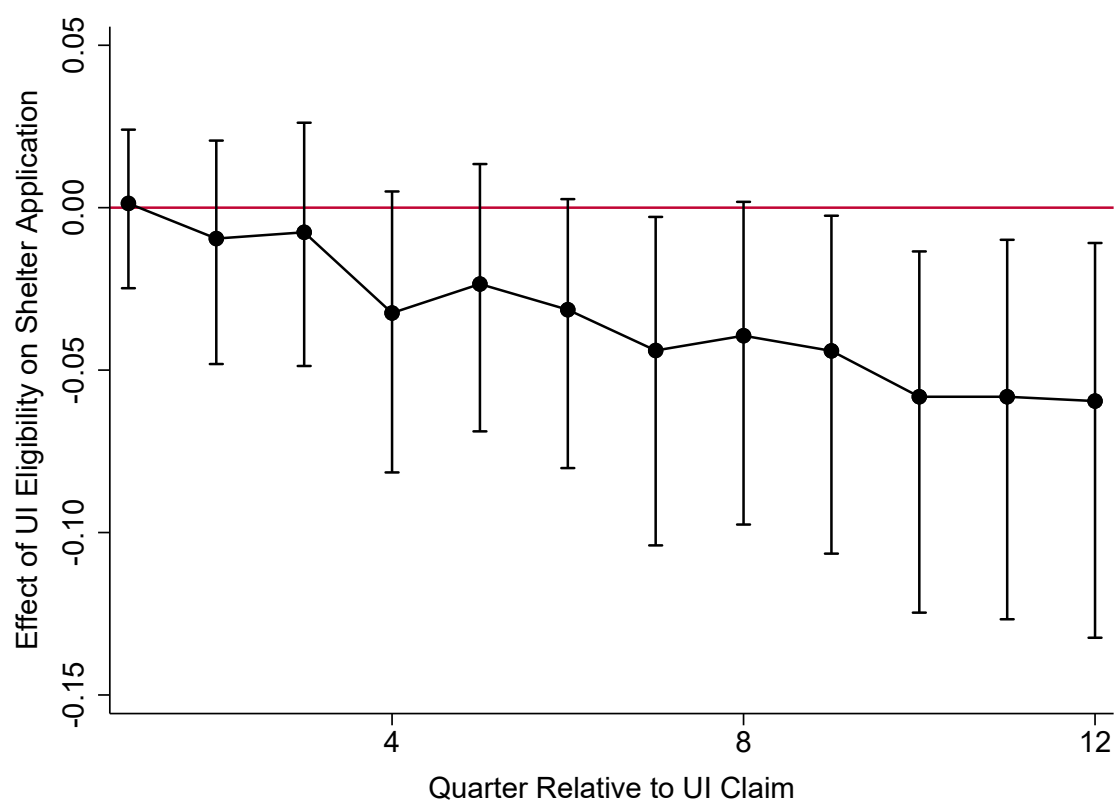
Notes: Panel a (b) plots the proportion of UI claimants eligible for UI (the mean benefit amount received). See Figure 3 for more information on figure construction.

Figure 5: Homeless Shelter Application within 36 Months



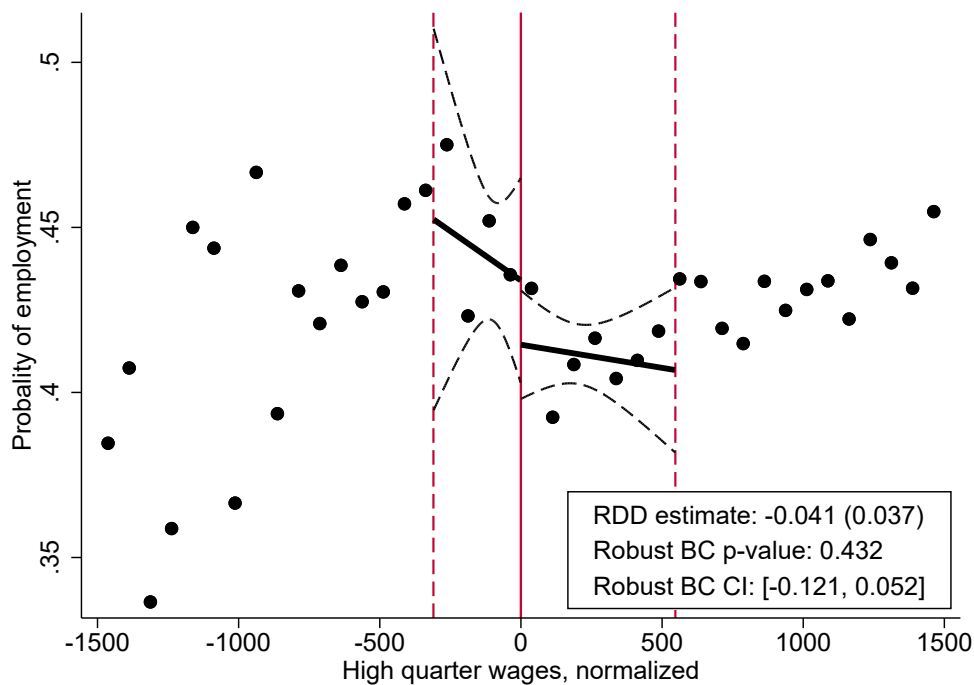
Notes: This figure plots the proportion of UI claimants who apply to stay in a homeless shelter within 36 months. The RDD estimate reported is scaled by UI eligibility. See Figure 3 for more information on figure construction.

Figure 6: Homeless Shelter Application through Time (Cumulative)

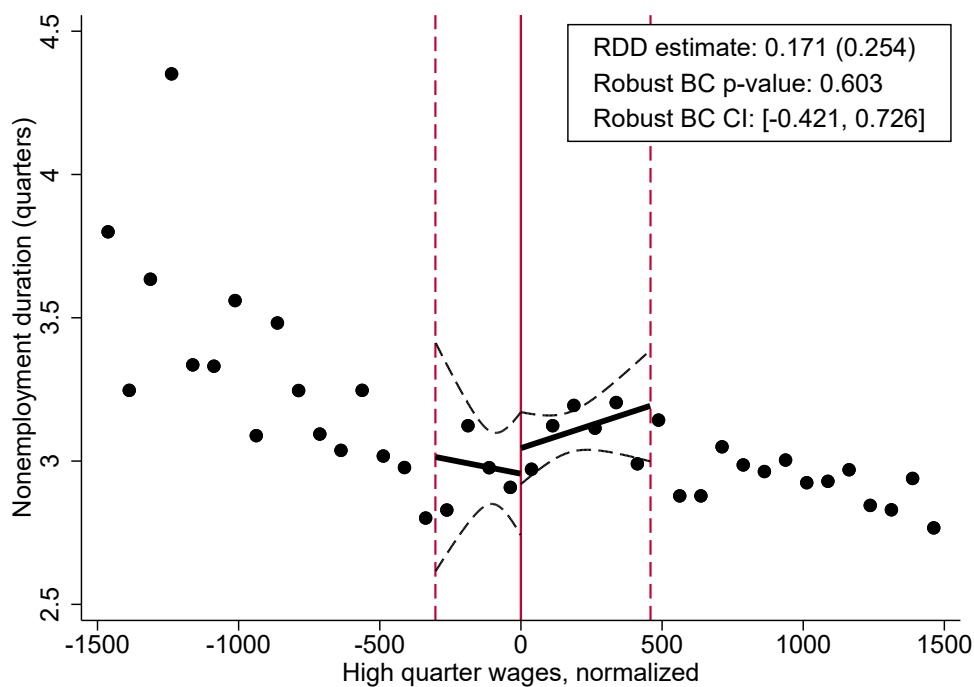


Notes: This figure plots estimates of the impact of UI eligibility on shelter applications within one quarter, two quarters, ... up to twelve quarters using a separate regression discontinuity for each quarter. The 95 percent confidence intervals are robust bias-corrected.

Figure 7: Nonemployment Spell



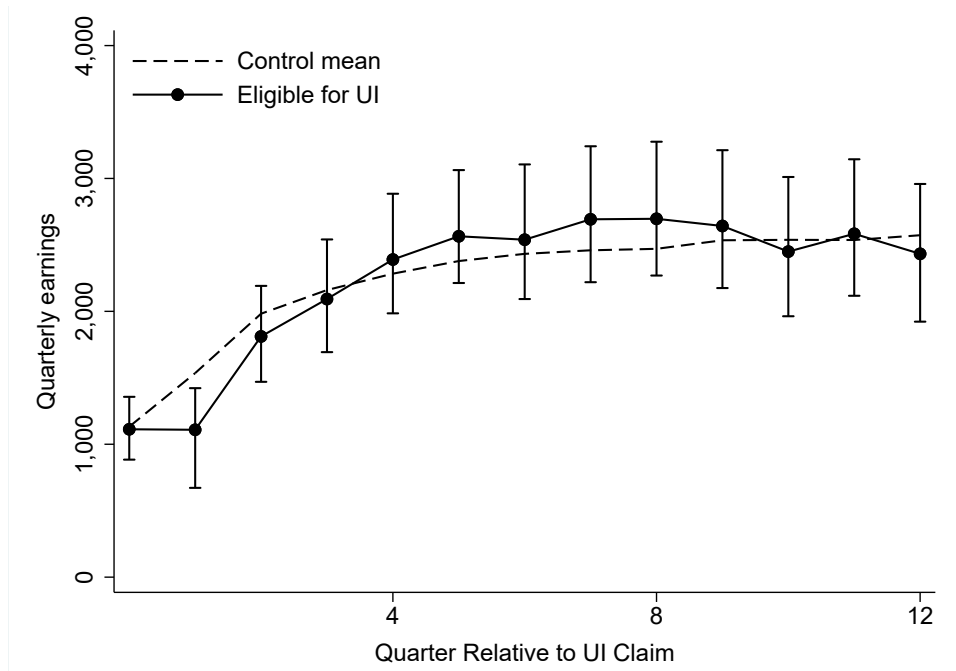
(a) Probability of employment in the quarter after UI claim



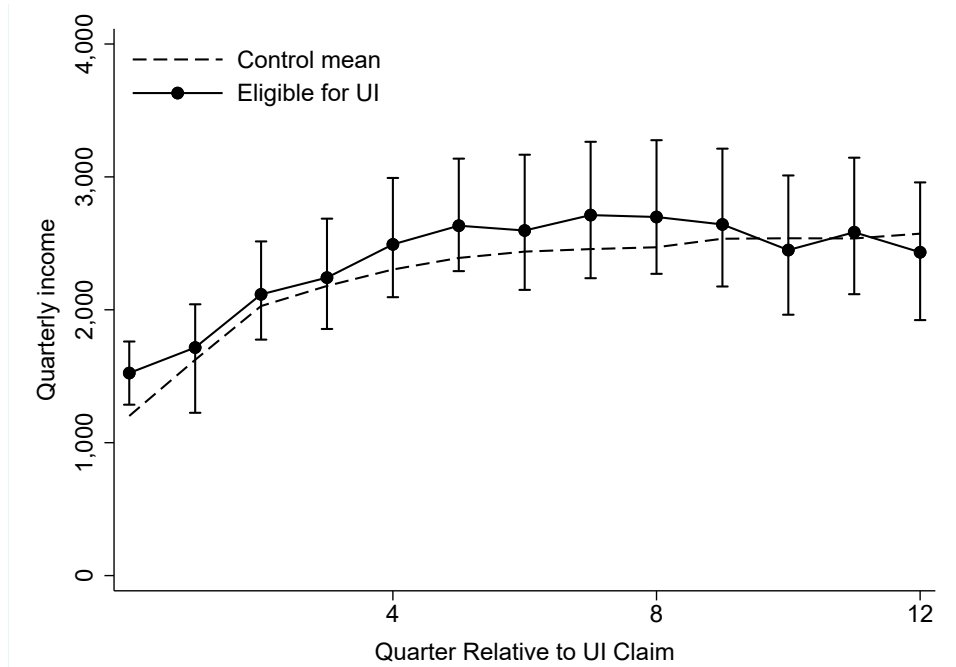
(b) Nonemployment duration

Notes: Panel a (b) plots the proportion of UI claimants employed in the quarter after the claim (nonemployment duration in quarters). The RDD estimates reported are scaled by UI eligibility. See Figure 3 for more information on figure construction.

Figure 8: Earnings and Income through Time



(a) Quarterly Earnings



(b) Quarterly Income (Earnings and UI)

Notes: These figures plot estimates of the impact of UI eligibility on earnings (panel a) and income (panel b) for each quarter relative to claim date using a separate regression discontinuity for each quarter. The dotted line indicates the control mean. The point estimates and confidence intervals are added to the control mean. The 95 percent confidence intervals are robust bias-corrected.

8 Tables

Table 1: Descriptive Statistics

	All (1)	Non-Courts (2)	Main (3)	Main, non-Courts (4)	Main, re-weighted (5)
<i>Demographic Characteristics</i>					
Female (%)	59.2	49.2	69.5	59.5	63.7
Age	35.21	35.78	30.33	32.02	31.32
Education					
Less than high school (%)	24.5	23.3	31.4	30.7	31.0
High school (%)	29.0	28.9	30.8	30.6	30.7
Some college (%)	33.2	31.8	30.6	29.8	30.1
Bachelor's degree (%)	8.9	10.8	4.9	6.2	5.7
Race and Ethnicity					
White (%)	16.9	22.7	13.2	17.9	16.0
Black (%)	38.9	29.6	44.2	33.6	38.0
Asian (%)	2.5	5.6	2.7	7.5	5.5
Hispanic (%)	35.6	34.3	34.5	31.6	32.8
<i>Employment, program participation, and housing stability before layoff</i>					
Previous year earnings	26,523	28,286	7,081	7,198	7,150
Ever claimed UI (%)	29.4	26.4	20.2	17.8	18.8
Received TANF (%)	21.4	14.0	33.0	20.8	26.8
Applied to shelter (since 2003) (%)	12.7	5.9	20.3	10.0	14.3
<i>Employment and housing stability after layoff</i>					
Entitled to UI (%)	79.7	74.1	66.5	60.1	62.8
UI receipt (%)	77.5	71.0	64.1	57.0	60.0
Monetarily eligible for UI (%)	98.4	98.2	87.0	84.5	85.5
Benefit amount (conditional on eligibility)	9,142	9,210	3,277	3,192	3,230
Weeks on UI (conditional on eligibility)	33.7	32.65	29.72	28.88	29.25
Employed quarter after UI claim (%)	43.5	42.9	42.9	41.1	41.8
Number of consecutive jobless quarters	2.96	3.1	2.98	3.23	3.13
Applied to shelter within 3 years after UI claim (%)	6.9	3.5	11.6	6.6	8.7
Observations	442,830	108,069	53,549	12,229	53,549

Notes: The sample for column 1 consists of all “new” UI claims (e.i., first claims or claims filed at least two years after a previous claim) for individuals in both the *courts* and *non-courts* sample from 2004-2016 and whose base period earnings are greater than 1.5 the high quarter earnings. The sample in column 2 contains a similar sample but only for the *non-courts* sample. Columns 3, 4, and 5 restrict the sample to contain claims within \$1500 of the highest-earning quarter threshold. Column 4 includes only claims from the *non-courts* sample. Column 5 up weights the random subsample of *non-courts* claims to better approximate the universe of claims from means-tested benefit recipients.

Table 2: Local Linear Estimates of Discontinuities in Predetermined Covariates

Covariate	Pt. est.	SE	P-Value	Control Mean	Percent Break
Pred. prob. of shelter app.	-0.001	(0.003)	[0.851]	0.100	-0.6
Pred. prob. emp.	0.001	(0.002)	[0.718]	0.426	0.2
<i>Demographic Characteristics</i>					
Female	-0.002	(0.017)	[0.921]	0.707	-0.3
Age	0.134	(0.480)	[0.989]	29.580	0.5
Education					
Less than high school	0.006	(0.016)	[0.790]	0.306	1.9
High school	0.004	(0.017)	[0.772]	0.306	1.2
Some college	-0.007	(0.018)	[0.717]	0.307	-2.2
Bachelor's degree	-0.002	(0.008)	[0.967]	0.057	-2.9
Race and ethnicity					
White	0.022	(0.013)	[0.103]	0.131	16.5
Black	0.023	(0.021)	[0.268]	0.471	4.9
Asian	-0.005	(0.006)	[0.401]	0.027	-19.9
Hispanic	-0.003	(0.018)	[0.947]	0.311	-1.0
<i>Homelessness and employment before layoff</i>					
Prior shelter app.	-0.006	(0.016)	[0.725]	0.213	-2.8
Quarterly earnings (two years before)	-171.01	(135.34)	[0.256]	2,205.02	-7.8
Employed (two years before)	0.004	(0.014)	[0.706]	0.366	1.2
Ever Claimed UI	0.008	(0.017)	[0.442]	0.208	3.8
Recieved TANF	-0.004	(0.023)	[0.709]	0.327	-1.2

Notes: Each row corresponds to an estimate of the discontinuity using equation (1) with the listed covariate as the outcome variable. Bandwidths are selected using the method proposed by Calonico et al. (2014) allowing bandwidths to vary on either side of the threshold. The Pt. est. reports $\hat{\beta}_y$. The SE column reports the convectional heteroskedastic-robust standard error. The P-Value column reports the robust bias-corrected p-value. The Percent Break column gives the linear discontinuity estimate as a percent of the control mean, calculate by taking the mean from the lower bandwidth up to the threshold. Predicted probability of shelter application within 36 months and employment in the quarter after layoff is calculated using a linear probability model with demographic information, prior earnings and employment, and prior homelessness. Quarterly earnings (two years before) is the average earnings from 6-9 quarters before the UI claim. Employed (two years before) is the fraction of quarters 6-9 quarters before the UI claim the claimant is has positive earnings.

Table 3: RDD Estimates of the UI Receipt and Benefit Amount

	Eligible (1)	Receipt (2)	Benefit Amount (3)	Weeks (4)
Estimate	0.575 (0.017)	0.548 (0.018)	1,275 (87.42)	18.10 (0.65)
Control mean	0.099	0.096	421	2.73
Confidence interval	[0.527, 0.604]	[0.499, 0.577]	[1,073, 1,479]	[16.46, 19.52]
Bandwidth	- 291, 399	- 272, 406	- 395, 765	- 364, 836
Observations	10,511	10,533	21,160	23,258

Notes: Each column presents estimates of β_Y from equation (1) where the outcome variable, Y_i , in column 1 is whether the claim is deemed eligible, in column 2 whether the claimant received UI benefits, in column 3 is the total benefit amount received, and in column 4 is the number of weeks a claimant receives UI. Bandwidths are selected using the method proposed by Calonico et al. (2014) allowing bandwidths to vary on either side of the threshold. Conventional heteroskedastic-robust standard errors are included below point estimate in parentheses. The control mean is calculate by taking the mean from the lower bandwidth up to the threshold. The 95 percent confidence intervals are robust bias-corrected.

Table 4: RDD Estimates of UI Eligibility on Homeless Shelter Application

	One Year		Two Years		Three Years	
	(1)	(2)	(3)	(4)	(5)	(6)
Estimate	-0.033 (0.020)	-0.032 (0.019)	-0.040 (0.023)	-0.039 (0.022)	-0.060 (0.027)	-0.060 (0.026)
Control mean	0.056	0.056	0.095	0.095	0.131	0.131
First stage	0.581	0.580	0.592	0.593	0.587	0.587
Confidence interval	[-0.082, 0.006]	[-0.082, 0.005]	[-0.099, 0.005]	[-0.098, 0.002]	[-0.135, -0.009]	[-0.132, -0.011]
P-value	0.092	0.083	0.074	0.059	0.024	0.021
Bandwidth	- 290, 556	- 298, 558	- 305, 731	- 309, 796	- 320, 705	- 320, 733
Observations	14,044	14,166	18,152	20,100	16,239	16,904

Notes: Each column presents estimates of β_y/β_d from equation (1) and (2) where the endogenous variable, D_i , is whether a claimant is deemed eligible. The outcome variable is whether a claimant applied to stay at a homeless shelter within one year in columns 1-2, within two years in columns 3-4, and within three years in columns 5-6. Even columns include controls such as age, race, education, prior homelessness, prior earnings and prior employment. Bandwidths are selected using the method proposed by Calonico et al. (2014) allowing bandwidths to vary on either side of the threshold. Conventional heteroskedastic-robust standard errors are included below point estimate in parentheses. The control mean is calculate by taking the mean from the lower bandwidth up to the threshold. The p-values and 95 percent confidence intervals are robust bias-corrected.

Table 5: RDD Estimates of UI Eligibility on Labor Supply

	Employed Quarter after Claim (1)	Nonemployment Duration (2)
Estimate	-0.041 (0.037)	0.171 (0.254)
Control mean	0.448	2.954
First stage	0.576	0.573
Confidence interval	[-0.121, 0.052]	[-0.421, 0.726]
P-value	0.432	0.603
Bandwidth	- 311, 548	- 304, 460
Observations	14,319	12,028

Notes: Each column presents estimates of β_{UI} from equation (2) where the endogenous variable is whether a claimant is deemed eligible. The outcome variable is whether a claimant is employed in the quarter after the claim is filed column 1 and nonemployment duration (quarters) in columns 2. Estimates include controls such as age, race, education, prior homelessness, prior earnings and prior employment. Bandwidths are selected using the method proposed by Calonico et al. (2014) allowing bandwidths to vary on either side of the threshold. Conventional heteroskedastic-robust standard errors are included below point estimate in parentheses. The control mean is calculate by taking the mean from the lower bandwidth up to the threshold. The p-values and 95 percent confidence intervals are robust bias-corrected.

Table 6: Robustness Checks

	Donuts					Same Bandwidth			Polynomial degree				
	Preferred (1)	10 (2)	25 (3)	50 (4)	100 (5)	200 (6)	Difference (7)	Sum (8)	Uniform Ker. (9)	Quadratic (10)	Cubic (11)	Quartic (12)	Reweightd (13)
<i>Homelessness</i>													
Shelter app. (3 years)	-0.060 (0.027) [0.024]	-0.056 (0.028) [0.046]	-0.061 (0.032) [0.063]	-0.032 (0.041) [0.466]	-0.082 (0.062) [0.277]	0.012 (0.073) [0.949]	-0.047 (0.030) [0.146]	-0.046 (0.030) [0.157]	-0.063 (0.026) [0.017]	-0.067 (0.032) [0.037]	-0.065 (0.037) [0.100]	-0.063 (0.037) [0.103]	-0.073 (0.030) [0.015]
Shelter app. (2 years)	-0.040 (0.023) [0.074]	-0.023 (0.021) [0.300]	-0.040 (0.024) [0.095]	-0.033 (0.031) [0.293]	-0.083 (0.047) [0.101]	0.003 (0.061) [0.877]	-0.021 (0.027) [0.447]	-0.027 (0.025) [0.305]	-0.036 (0.024) [0.092]	-0.039 (0.030) [0.240]	-0.035 (0.032) [0.354]	-0.039 (0.031) [0.264]	-0.063 (0.025) [0.013]
Shelter app. (1 years)	-0.033 (0.020) [0.092]	-0.029 (0.022) [0.183]	-0.029 (0.023) [0.197]	-0.011 (0.021) [0.663]	-0.045 (0.033) [0.191]	0.024 (0.044) [0.681]	-0.023 (0.022) [0.262]	-0.025 (0.021) [0.216]	-0.030 (0.018) [0.102]	-0.046 (0.023) [0.050]	-0.038 (0.026) [0.161]	-0.036 (0.025) [0.200]	-0.056 (0.024) [0.017]
<i>Labor Supply</i>													
Employed quarter after claim	-0.040 (0.037) [0.486]	-0.016 (0.038) [0.986]	-0.054 (0.047) [0.399]	-0.109 (0.055) [0.109]	-0.029 (0.065) [0.844]	-0.132 (0.065) [0.099]	-0.032 (0.040) [0.622]	-0.034 (0.039) [0.612]	-0.032 (0.034) [0.520]	-0.030 (0.047) [0.634]	-0.025 (0.049) [0.736]	-0.030 (0.048) [0.631]	-0.058 (0.046) [0.208]
Nonemployment Duration	0.136 (0.263) [0.731]	0.156 (0.249) [0.694]	0.085 (0.311) [0.936]	0.886 (0.418) [0.057]	0.008 (0.430) [0.892]	1.603 (0.578) [0.011]	0.093 (0.278) [0.879]	0.125 (0.260) [0.831]	0.288 (0.242) [0.379]	0.099 (0.321) [0.843]	0.059 (0.331) [0.969]	0.157 (0.323) [0.688]	0.508 (0.286) [0.076]
<i>First Stage</i>													
Eligible for UI	0.575 (0.017) [0.000]	0.586 (0.017) [0.000]	0.579 (0.021) [0.000]	0.573 (0.027) [0.000]	0.600 (0.027) [0.000]	0.641 (0.038) [0.000]	0.576 (0.018) [0.000]	0.576 (0.016) [0.000]	0.580 (0.015) [0.000]	0.562 (0.019) [0.000]	0.561 (0.020) [0.000]	0.565 (0.020) [0.000]	0.509 (0.024) [0.000]
Benefit Amount	1274.814 (87.417) [0.000]	1314.824 (88.076) [0.000]	1316.654 (91.736) [0.000]	1352.606 (125.551) [0.000]	1309.508 (166.864) [0.000]	1090.150 (282.834) [0.006]	1226.597 (97.128) [0.000]	1228.600 (89.556) [0.000]	1284.534 (82.777) [0.000]	1264.087 (116.469) [0.000]	1226.463 (132.303) [0.000]	1256.454 (133.566) [0.000]	1254.805 (114.194) [0.000]
<i>Balance</i>													
Pred. prob. of shelter app.	-0.001 (0.003) [0.851]	0.001 (0.004) [0.666]	-0.001 (0.004) [0.751]	0.004 (0.004) [0.790]	-0.002 (0.005) [0.443]	0.002 (0.008) [0.383]	0.000 (0.003) [0.861]	-0.001 (0.003) [0.777]	0.000 (0.003) [0.438]	-0.001 (0.004) [0.968]	-0.001 (0.004) [0.899]	-0.002 (0.004) [0.970]	-0.002 (0.004) [0.865]
Pred. prob. emp.	0.001 (0.002) [0.718]	0.000 (0.003) [0.798]	0.001 (0.003) [0.751]	0.001 (0.004) [0.790]	0.005 (0.006) [0.443]	-0.005 (0.005) [0.383]	0.000 (0.002) [0.861]	0.000 (0.003) [0.777]	0.002 (0.002) [0.438]	0.000 (0.003) [0.968]	0.001 (0.003) [0.899]	0.000 (0.003) [0.970]	0.001 (0.003) [0.865]
Prior shelter app.	-0.006 (0.016) [0.725]	0.006 (0.017) [0.696]	-0.010 (0.019) [0.602]	0.012 (0.021) [0.497]	-0.013 (0.027) [0.764]	0.023 (0.041) [0.502]	-0.003 (0.017) [0.876]	-0.003 (0.003) [0.904]	-0.001 (0.015) [0.957]	-0.011 (0.020) [0.594]	-0.010 (0.022) [0.661]	-0.010 (0.022) [0.591]	-0.008 (0.017) [0.638]
Quarterly earnings (two years before)	-171.010 (135.343) [0.256]	-227.661 (150.935) [0.144]	-187.253 (170.474) [0.327]	-176.043 (200.048) [0.420]	1031.649 (416.950) [0.011]	-643.175 (468.225) [0.140]	-197.747 (139.545) [0.163]	-197.475 (139.262) [0.174]	-155.121 (149.841) [0.298]	-216.170 (186.241) [0.322]	-257.293 (219.084) [0.236]	-229.136 (217.691) [0.276]	-181.605 (184.884) [0.326]
Observations	16,239	15,687	13,115	10,334	9,435	27,516	8,033	7,793	14,139	28,770	34,170	57,818	.

Notes: Each column presents RDD estimates at the high quarter threshold using equations (1) and/or (2). The outcome variable is shown in the first column. The *homelessness* and *Labor Supply* sections are scaled by the UI eligibility, while the remaining columns are reduced-form estimates. Columns 1 reports our preferred specification. Columns 2-6 reports estimates omitting observations near the threshold. Columns 7 and 8 reports estimates using the same bandwidth on either side of the cutoff with column 7 (using) the bandwidth mean-square optimal using the difference (sum) between the left and right point estimates. Column 9 uses a uniform kernel. Columns 10-12 use higher order polynomials. Column 13 reweights the sample to account for individuals with an eviction filing being over represented in our sample. Because bandwidths change across outcome variables, observation reports the number of observation with shelter app. within 3 years as the outcome variable. Bandwidths are selected using the method proposed by Calonico et al. (2014) allowing bandwidths to vary on either side of the threshold. Conventional heteroskedastic-robust standard errors are included below point estimate in parentheses. The p-values are robust bias-corrected, and in brackets

Table 7: RDD Estimates Using Alternative Endogenous Variables

	Receipt (1)	Benefit Amount (thousands) (2)
<i>Homelessness</i>		
Shelter app. (3 years)	-0.064 (0.028) [0.023]	-0.023 (0.011) [0.049]
Shelter app. (2 years)	-0.041 (0.024) [0.077]	-0.016 (0.010) [0.104]
Shelter app. (1 years)	-0.034 (0.020) [0.086]	-0.016 (0.008) [0.037]
<i>Labor Supply</i>		
Employed quarter after claim	-0.045 (0.040) [0.437]	-0.020 (0.015) [0.344]
Nonemployment Duration	0.161 (0.277) [0.678]	0.118 (0.092) [0.354]
Observations	16,778	20,622

Notes: Each column presents estimates of β_{UI} from equation (2) where the endogenous variable is whether the claimant receives UI in column 1 and the benefit amount received in column 2. The outcome variable is shown in the first column. Because bandwidths change across outcome variables, observation reports the number of observation with shelter app. within 3 years as the outcome variable. Bandwidths are selected using the method proposed by Calonico et al. (2014) allowing bandwidths to vary on either side of the threshold. Conventional heteroskedastic-robust standard errors are included below point estimate in parentheses. The p-values are robust bias-corrected. and in brackets

Table 8: RDD Estimates Using Alternative Thresholds

	Base Period (1)	Joint (2)
<i>Homelessness</i>		
Shelter app. (3 years)	0.013 (0.022) [0.530]	-0.016 (0.018) [0.450]
Shelter app. (2 years)	0.022 (0.019) [0.353]	-0.004 (0.015) [0.818]
Shelter app. (1 years)	0.002 (0.014) [0.746]	-0.008 (0.012) [0.620]
<i>Labor Supply</i>		
Employed quarter after claim	-0.058 (0.033) [0.123]	-0.051 (0.027) [0.103]
Nonemployment Duration	0.250 (0.210) [0.342]	0.205 (0.189) [0.397]
<i>First Stage</i>		
Eligible for UI	0.453 (0.013) [0.000]	0.492 (0.013) [0.000]
Benefit Amount	2817.931 (151.649) [0.000]	2092.303 (106.416) [0.000]
<i>Balance</i>		
Pred. prob. of shelter app.	0.000 (0.002) [0.652]	0.000 (0.002) [0.872]
Pred. prob. emp.	0.000 (0.002) [0.920]	0.001 (0.002) [0.605]
Prior shelter app.	0.002 (0.012) [0.848]	-0.001 (0.009) [0.838]
Quarterly earnings (two years before)	-195.133 (197.768) [0.520]	-281.044 (160.073) [0.118]
Observations	24,884	29,115

Notes: Each column presents RDD estimates at the high quarter threshold using equations (1) and/or (2) but exploiting different thresholds. The outcome variable is shown in the first column. The *homelessness* and *Labor Supply* sections are scaled by the UI eligibility, while the remaining columns are reduced-form estimates. Columns 1 uses the base period earnings threshold which is 1.5 time the highest earning quarter. Column 2 combines both thresholds by taking the minimum of the two running variables, high quarter earnings minus the cut off and base period earnings minus 1.5 times high quarter earnings, as the new running variable. Because bandwidths change across outcome variables, observation reports the number of observation with shelter app. within 3 years as the outcome variable. Bandwidths are selected using the method proposed by Calonico et al. (2014) allowing bandwidths to vary on either side of the threshold. Conventional heteroskedastic-robust standard errors are included below point estimate in parentheses. The p-values are robust bias-corrected. and in brackets